

# A study of Mach-Zehnder interferrometer design, fabrication and analysis.

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**Abstract:** The integrated photonics Mach-zehnder interferrometer (MZI) is a foundational and versatile building block, employed on chips in various applications ranging from telecommunications to biosensing. For each application, the MZI circuit is designed, tailored and fabricated to meet functional needs. This study presents a full design, layout, fabrication, and analysis cycle for realization of reliable MZI devices with expected performance metrics.

## 1. Introduction

The past several decades have witnessed an exponential growth in the development and maturation of integrated photonics-based technology. Integrated photonics has been adopted as the key enabling technology in a myriad of applications within fields ranging from telecommunications [1], quantum computing [2], to various forms of metrology including but not limited to strain [3,4], temperature [5], pressure [6], shape- and even biosensing [7]. The Mach-Zehnder interferrometer (MZI) circuit has served as a foundational building block on chips designed for several of these applications, acting as an optical switch [8], interrogator building block [9], acoustical sensor [10], biosensor [11] [12], and more. In each of the above, the MZI circuit is designed and tailored to the specific application starting from waveguide cross-section design and spectral behavior to the complete circuit design, including the choice of splitting the building block and routing of the separate waveguide branches. In addition, deviations due to fabrication tolerances should also be taken into account through corner analysis studies to arrive at an MZI circuit with key performance indicators within desired upper and lower specification limits, USL and LSL, respectively.

The study presented in this contribution details one full cycle of a typical design, modeling, layout and fabrication cycle of several MZI circuits. Firstly, a theoretical model of the MZI transmission function is established as a basis for the circuit design. Then the waveguide design is presented with numerical modeling of the fundamental mode for the selected waveguide geometry. This includes computations of the spectral behaviour of the waveguide effective refractive index,  $n_{\text{eff}}$ , and group index,  $n_g$  around the target operating wavelength to arrive at a compact waveguide model for subsequent circuit-level MZI design studies. The fabrication variations are considered with a corner analysis taking into account by repeating the above numerical study for several configurations of waveguide width and thickness and establishing respective compact models. These compact models are employed in circuit level numerical studies to yield MZI circuit performance with respective upper and lower bounds resulting from fabrication variations. The results and design parameters obtained from the waveguide and circuit level numerical study serve as a basis for the MZI design layout submitted for fabrication. Fabrication is carried out through an e-beam lithographic process and the devices are tested with an automated photonic integrated circuit testing apparatus. The test results are then analyzed to ultimately yield experimental  $n_{\text{eff}}$  and  $n_g$  values within the upper and lower limits of what was obtained through numerical models.

## 2. Theory

The MZI circuit fundamentally comprises two splitters and two waveguide arms, where the waveguide arms can be designed to be symmetrical or not. In the prior case, the MZI is balanced

46 while in the latter case the MZI is unbalanced which is illustrated schematically in Fig 1.

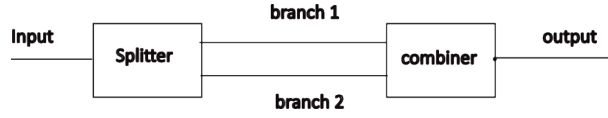


Fig. 1. Schematic representation of MZI circuit

47 An analytical expression to model the MZI transfer function can be derived considering  
 48 the optical functionality of each circuit component. Firstly, the splitter components can be  
 49 chosen from a list of several building blocks including but not limited to y-branches, directional  
 50 couplers, and MMIs. The traditional y-branch splitter, considered for the purposes of this article,  
 51 redistributes the incoming optical intensity,  $I_i$ , equally across the two branches. The electric  
 52 fields in the two branches can be expressed as

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}} \quad (1)$$

53 where  $E_1$ ,  $E_2$  and  $E_i$  are the electric fields in branches 1 and 2 and the incident electric field,  
 54 respectively. Assuming plane-wave propagation, the electric fields  $E_{o,1}$  and  $E_{o,2}$  at the output of  
 55 waveguide branches 1 and 2, respectively, can be expressed as

$$E_{o,m} = E_m e^{-j\beta_m L_m - \frac{\alpha_m}{2} L_m}, \quad m = 1, 2 \quad (2)$$

56 where  $L_m$  and  $\alpha_m$  are the lengths and propagation losses of the waveguides in branch  $m$ . The  
 57 propagation constant  $\beta_m$  dictates the accumulated phase in waveguide branches  $m$  and can be  
 58 expressed as

$$\beta_m = \frac{2\pi n_{\text{eff},m}}{\lambda}, \quad m = 1, 2 \quad (3)$$

59 where  $\lambda$  is the vacuum wavelength of light, and  $n_{\text{eff},m}$  the effective index of the waveguide in  
 60 branch  $m$ . Both  $n_{\text{eff},m}$  and  $\alpha_m$  are highly dependent on the waveguide material and geometry, the  
 61 electro-magnetic mode confinement, and the quality and tolerances of the fabrication. For the  
 62 purposes of the study presented in this article, the waveguides in both branches are identical and  
 63 only differing in their lengths  $L_1$  and  $L_2$ .

64 The final y-branch combines the electric fields  $E_{o,1}$  and  $E_{o,2}$  at the output of the two waveguide  
 65 branches with the resulting electric field,  $E_o$ , coupled to the output waveguide. This is expressed  
 66 as

$$E_o = \frac{E_{o,1} + E_{o,2}}{\sqrt{2}}. \quad (4)$$

67 Combining above Eqs. 1 - 4, assuming waveguide branches with identical waveguides and  
 68 differing length  $L_1$  and  $L_2$ , and neglecting the propagation loss  $\alpha$  give the following expression  
 69 for  $E_o$

$$E_o = E_i \frac{e^{-j\beta L_1} + e^{-j\beta L_2}}{2}. \quad (5)$$

70 Consequentially, the relationship between the input and output ports of the MZI circuit is  
 71 expressed as

$$I_o = \frac{I_i}{2} (1 + \cos(\beta \Delta L)) \quad (6)$$

72 where  $\Delta L = L_1 - L_2$  is the length difference between waveguide branches 1 and 2.

73 A fundamental property of an MZI circuit is the difference in accumulated phase,  $\phi$ , between  
74 the two waveguide arms, expressed as

$$\phi = \beta \Delta L = \frac{2\pi n_{\text{eff}} \Delta L}{\lambda}. \quad (7)$$

75 A shift in the accumulated phase,  $\Delta\phi$ , is caused by several factors including shifts in wavelength,  
76 temperature, and surrounding cladding material. Considering only a shift in wavelength,  $\Delta\lambda$ , the  
77 shift in phase can be expressed as

$$\Delta\phi = \beta \Delta L = \left( \frac{1}{\lambda} \frac{dn_{\text{eff}}}{d\lambda} - \frac{n_{\text{eff}}}{\lambda^2} \right) 2\pi \Delta L \Delta\lambda = \left( \lambda \frac{dn_{\text{eff}}}{d\lambda} - n_{\text{eff}} \right) \frac{2\pi \Delta L}{\lambda^2} \Delta\lambda. \quad (8)$$

78 The wavelength shift for which  $\Delta\phi = 2\pi$  is called the Free Spectral Range (FSR), for which the  
79 expression follows from Eqn 8 to be

$$\text{FSR} = \frac{\lambda^2}{\left( n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) \Delta L} = \frac{\lambda^2}{n_g \Delta L} \quad (9)$$

80 where the group index,  $n_g$ , accounts for the dispersion to the first order. The FSR will be a  
81 principle design parameter which is considered in the study presented in this article.

### 82 3. Modeling and simulation

#### 83 3.1. The waveguide

84 As mentioned in Section 2, the effective index,  $n_{\text{eff}}$ , and consequentially the group index,  $n_g$ ,  
85 are highly dependent on waveguide geometry and core and cladding materials. In this study,  
86 silicon-on-insulator is employed as the material platform with the standard stripe waveguide as  
87 the fundamental waveguide geometry. The width and thickness of the waveguide are  $w = 0.5 \mu\text{m}$   
88 and  $t = 0.22 \mu\text{m}$ , per industry standard, and a spectral operational bandwidth of 100 nm centered  
89 around  $\lambda = 1550 \text{ nm}$ . Besides, the waveguide core material is silicon (Si) while the surrounding  
90 cladding is ( $\text{SiO}_2$ ).

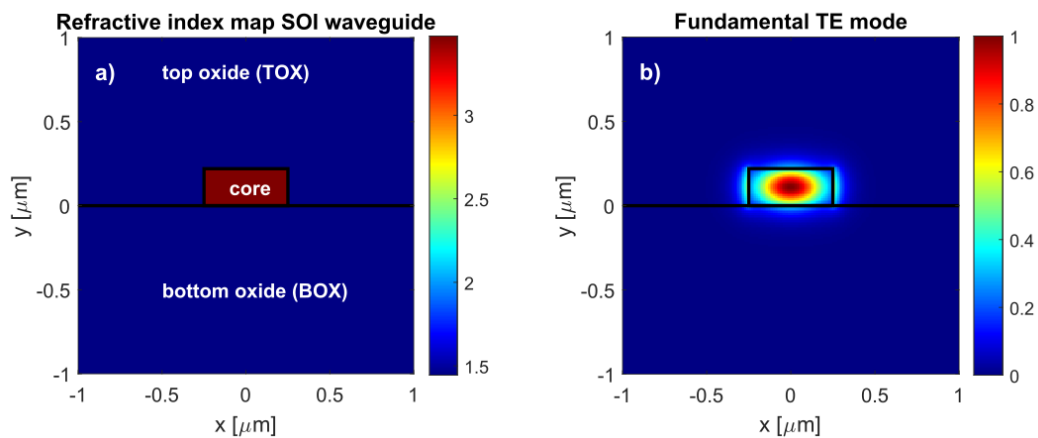


Fig. 2. Normalized intensity color map of the fundamental TE-mode for an industry standard SOI waveguide with width and thickness of  $w = 0.5 \mu\text{m}$  and  $t = 0.22 \mu\text{m}$ , respectively. The waveguide contour is highlighted with a solid black line.

With Lumerical MODE, a modal analysis simulation has been carried out for the chosen geometry, waveguide material and operational optical bandwidth to yield three supported modes: the fundamental TE and TM modes, and the first order TE mode. Modeling of the MZI circuit will be carried out for the fundamental TE mode as shown in Fig 2 at  $\lambda = 1550$  nm.

The effective index was subsequently simulated for the TE-mode and wavelength range of 1500 nm - 1600 nm to model the waveguide dispersion resulting from material dispersion and modal distribution. The waveguide dispersion results are expressed in terms of the effective and group indices and are shown in Figs 3 a) and b), respectively.

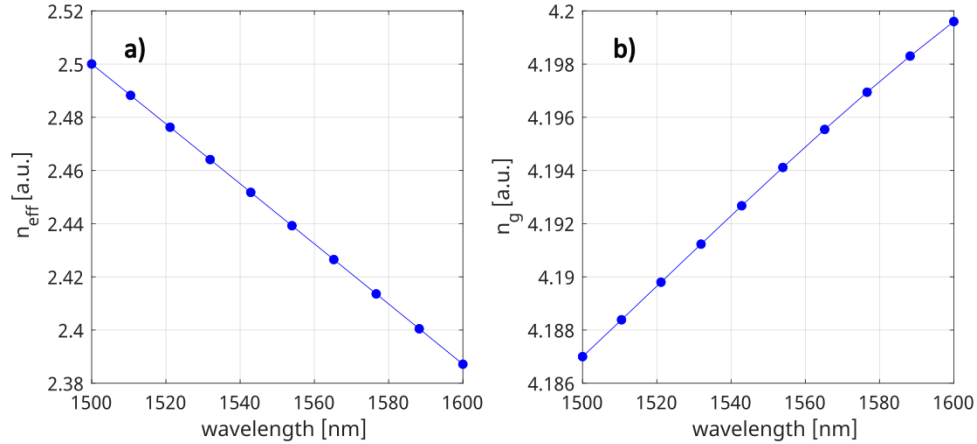


Fig. 3. Waveguide dispersion expressed as a) the effective index and b) the group index for a wavelength range of 1500 nm -1600 nm.

The waveguide dispersion data was then employed to construct a compact model for the waveguide, which can be expressed with the following truncated Taylor-expansion centered around 1550 nm,

$$n_{\text{eff}} = n_1 - n_2 \cdot (\lambda - 1.55) - n_3 \cdot (\lambda - 1.55)^2 \quad (10)$$

The compact model for a SOI-waveguide with  $w = 0.5 \mu\text{m}$  and  $t = 0.22 \mu\text{m}$  modeled above is:

$$n_{\text{eff}} = 2.444 - 1.129 \cdot (\lambda - 1.55) - 0.041 \cdot (\lambda - 1.55)^2 \quad (11)$$

where the wavelength is expressed in  $\mu\text{m}$ .

To account for variations in waveguide width and height characteristic for the Ebeam process used to fabricate the MZI circuits proposed in this study, the above simulations are repeated for each of the four remaining corner points schematically shown in Fig. 4a). The waveguide dispersion results resulting from this corner analysis are shown in Fig. 4b). The compact models for each waveguide height-width configuration pair, as shown in Fig. 4a), are stored for application in circuit simulations and modeling. Within this study, several unbalanced MZI circuit are proposed with varying length differences  $\Delta L$  as listed in Table 1 along with corresponding FSR values computed with Eqn. 9 and  $n_g$  values computed from the waveguide compact models with the equation:

$$n_{\text{eff}} = n_1 - n_2 \cdot (\lambda_0) \quad (12)$$

which can be obtained by combining Eqn. 10 and the expression for  $n_g$  in Eqn. 9 with  $\lambda_0 = 1.55 \mu\text{m}$ .

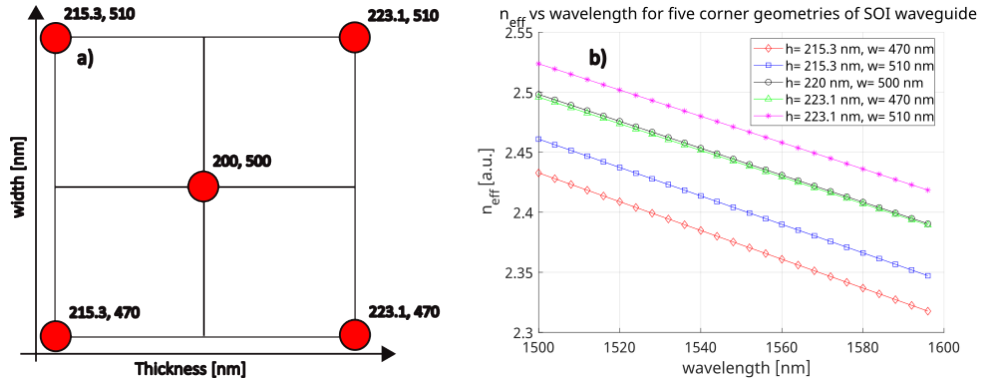


Fig. 4. Corner analysis overview with a) a schematic representation of the five waveguide (height, width) corner points investigated and b) the resulting waveguide dispersion results expressed in  $n_{\text{eff}}$  for a wavelength range of 1500 nm -1600 nm.

114 In addition, the FSR values include the minimum, nominal and maximum FSR values,  $\text{FSR}_{\text{min}}$ ,  
 115  $\text{FSR}_{\text{nom}}$  and  $\text{FSR}_{\text{max}}$ , respectively, resulting from the corner analysis.

Table 1. Selected values of  $\Delta L$  for proposed MZI circuits with corresponding computed minimum, nominal and maximum FSR values,  $\text{FSR}_{\text{min}}$ ,  $\text{FSR}_{\text{nom}}$ ,  $\text{FSR}_{\text{max}}$ , respectively

| $\Delta L [\mu\text{m}]$ | $\text{FSR}_{\text{min}} [\text{nm}]$ | $\text{FSR}_{\text{nom}} [\text{nm}]$ | $\text{FSR}_{\text{max}} [\text{nm}]$ |
|--------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| 25                       | 22.67                                 | 22.99                                 | 23.09                                 |
| 50                       | 11.33                                 | 11.49                                 | 11.55                                 |
| 100                      | 5.67                                  | 5.75                                  | 5.77                                  |
| 200                      | 2.83                                  | 2.87                                  | 2.89                                  |
| 400                      | 1.417                                 | 1.437                                 | 1.443                                 |

### 116 3.2. The MZI circuit

117 The MZI circuit was modelled using Lumerical Interconnect and comprise a y-branch splitter,  
 118 two waveguide branches and a y-branch combiner, according to the theoretical model presented  
 119 in section 2. Two grating couplers were included to allow for in/out coupling of light and wafer  
 120 level testing of modeled MZI circuit prototypes. Both Y-branches and grating couplers employed  
 121 in the modeled MZI circuits are taken from the EBeam compact model library included in the  
 122 SiEpic-EBeam-PDK.

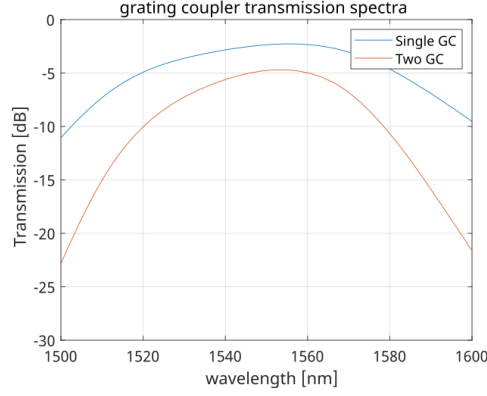


Fig. 5. The spectral transmission for a single grating coupler and a pair of grating couplers expressed in dB within the wavelength range of 1500 nm - 1600 nm.

123 The grating couplers will introduce wavelength dependent insertion loss to the MZI circuit which  
 124 will limit the operation bandwidth as is shown in Fig 5 a) and b) for a single and a pair of  
 125 grating couplers, respectively. The MZIs proposed in section 3.1 were all modeled in Lumerical  
 126 INTERCONNECT using the compact models for the waveguides described in section 3.1 and  
 127 the SiEpic-EBeam-PDK models for the grating couplers and Y-branch splitters. The expected  
 128 transmission spectra for selected MZI circuits is shown in Figs 6 a) - e).

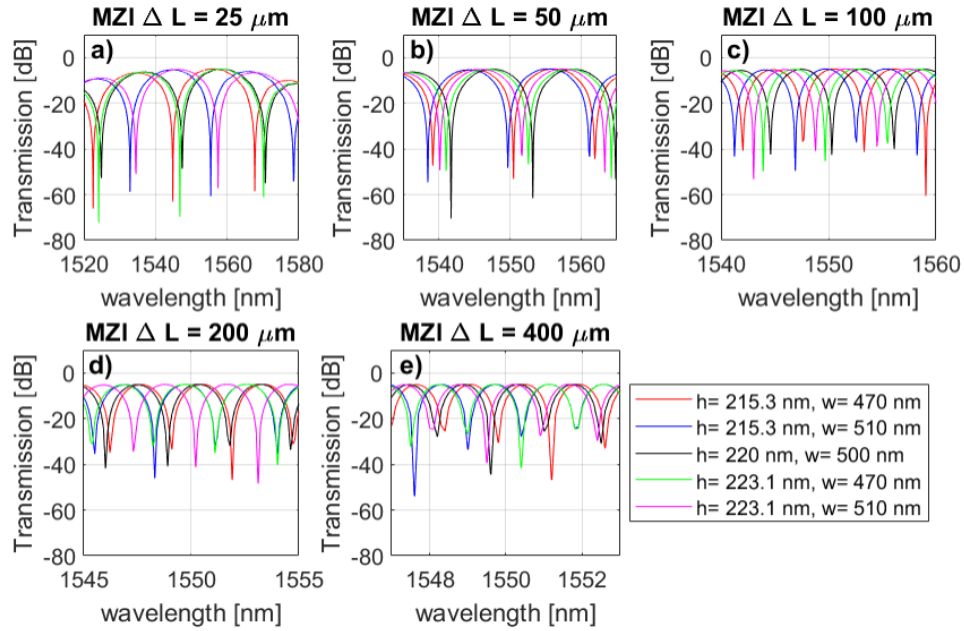


Fig. 6. The spectral transmission for proposed MZI circuits with  $\Delta L$  of a) 25  $\mu\text{m}$ , b) 50  $\mu\text{m}$ , c) 100  $\mu\text{m}$ , d) 200  $\mu\text{m}$  and e) 400  $\mu\text{m}$ . The MZI transmission plots transmission curves for the five corner waveguides shown in Fig. 4.

129 The FSR values  $\text{FSR}_{\min}$ ,  $\text{FSR}_{\max}$  and  $\text{FSR}_{\text{nom}}$ , calculated from the simulated spectra shown in  
 130 Figs. 6a)- e), agree with the values listed in table 1.

#### 4. Layout

The designs are implemented as Layout using KLayout with the SiEPIC-EBeam-PDK. A schematic overview of the primary floorplan with proposed design is shown in Figs 7 a) - g).

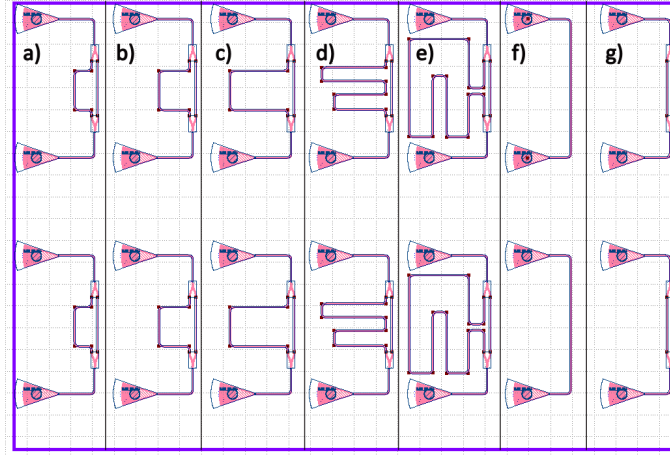


Fig. 7. Implemented MZI designs for fabrication including  $\Delta L$  values of a)  $25 \mu m$ , b)  $50 \mu m$ , c)  $100 \mu m$ , d)  $200 \mu m$  and e)  $400 \mu m$ . Besides the MZI designs, two calibration designs are implemented to determine the losses incurred by f) the grating couplers and g) the Y-branch splitters.

Five different MZI designs are implemented, corresponding to the five  $\Delta L$  values listed in Table 1. In addition, two calibration designs are included to determine and compensate for the insertion losses incurred by the grating couplers and the Y-branch splitters. There are two copies of each design to investigate the fabrication variations between designs. A secondary floorplan with additional designs is shown schematically in Figs 8 a) - g).

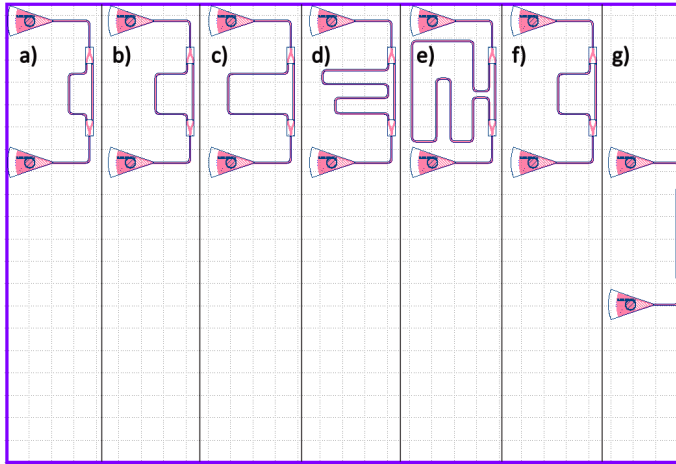


Fig. 8. Implemented MZI designs for fabrication including  $\Delta L$  values of a)  $25 \mu m$ , b)  $50 \mu m$ , c)  $100 \mu m$ , d)  $200 \mu m$  and e)  $400 \mu m$  and g) a calibration design with grating couplers and Y-branch splitters.

## 5. Fabrication

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2  $\mu\text{m}$  buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1  $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$ ) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2  $\mu\text{m}$  oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

## 6. Experimental data

To characterize the devices, a custom-built automated test setup [13] [14] with automated control software written in Python was used [15]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [16]. A 90° rotation was used to inject light into the TM grating couplers [16]. A polarization maintaining fibre array was used to couple light in/out of the chip [17].

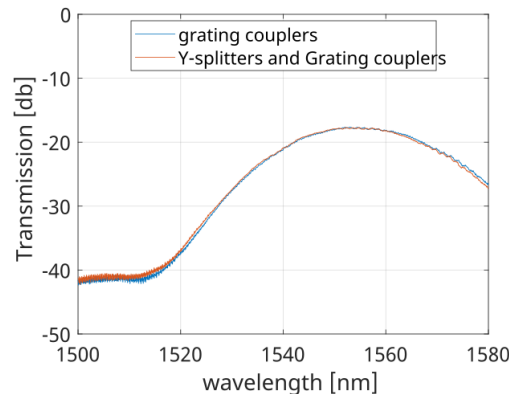


Fig. 9. The experimentally measured spectral transmission for a single grating coupler and a pair of grating couplers expressed in dB within the wavelength range of 1500 nm - 1600 nm.

170 The transmission spectra obtained from calibration structures Figs 7 f) - g) are shown in Fig. 9.  
 171 These results show a negligible loss contribution from the Y-splitter branch. Furthermore, both  
 172 form and magnitude of the spectral curve in Fig. 9 differ significantly from the model results  
 173 presented in Fig. 5. The experimentally obtained transmission curves both exhibit a higher  
 174 insertion loss and a narrower spectral transmission band. Discrepancy between the measured  
 175 transmission and simulated spectra can possibly originate from inaccuracies in the grating coupler  
 176 S-matrix or compact models. The chosen grating coupler models may have also been developed  
 177 according to fabrication methods and tolerances of another foundry. However, the exact root  
 178 cause behind this discrepancy is a topic for another investigation and out of scope for the current  
 179 study. Besides, the spectral transmission results from the calibration structures will be used to  
 180 normalize results obtained from the fabricated MZI structures.

## 181 7. Analysis

182 As briefly mentioned in section 6, the measured MZI transmission spectra are normalized by  
 183 spectra measured from the calibration structures, e.g. design in Fig. 7 f) or g) with spectra as  
 184 shown in Fig. 9. To avoid introduction of measurement noise from the calibration structures into  
 185 the normalized MZI data, the normalization is carried out with a polynomial-based fit curve of  
 186 the calibration structure transmission curves as shown in Figs 10 a) and b).

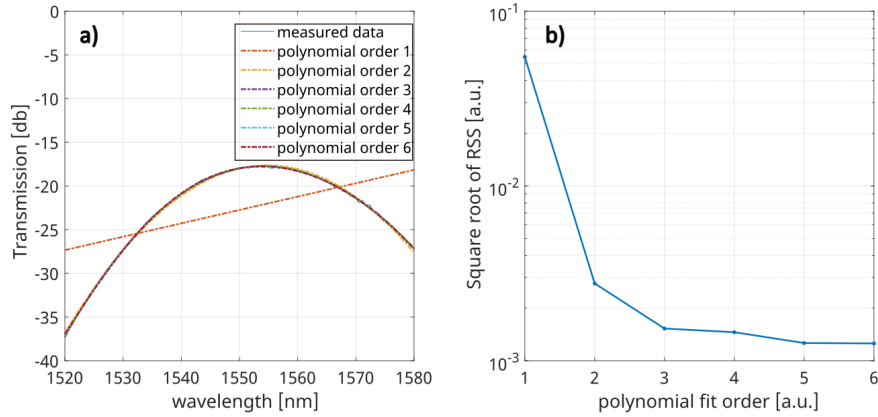


Fig. 10. a) Measured calibration design transmission spectra between 1520 nm and 1580 nm with polynomial fit curves of order 1 up to 6. The fit accuracy of each polynomial curve is measured by b) the square root of the sum of squared residuals (RSS).

187 For an accurate fit model, wavelength interval is limited to 1520 nm - 1580 nm and a polynomial  
 188 order of 5 is shown in Fig 10 b) to be most optimal. The normalized MZI measured data is shown  
 189 in Figs. 11a) - e).

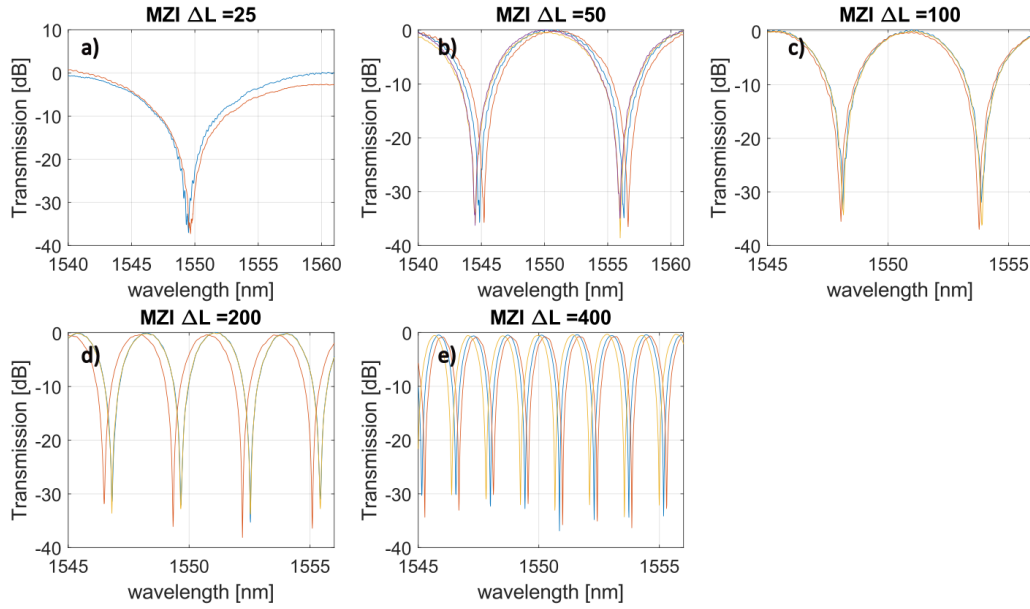


Fig. 11. The spectral transmission for experimentally measured MZI circuits with  $\Delta L$  of a) 25  $\mu\text{m}$ , b) 50  $\mu\text{m}$ , c) 100  $\mu\text{m}$ , d) 200  $\mu\text{m}$  and e) 400  $\mu\text{m}$ .

190 The experimentally determined FSR values are extracted from the normalized MZI transmission  
 191 curves by 1.) estimating the location of the sharply cusped minima through peak detection and 2.)  
 192 subsequent computation of the difference between estimated locations of neighboring minima.  
 193 The estimated FSR values for all measured MZI transmission curves are shown in Fig. 12 a).  
 194 The  $n_g$  values shown in Fig. 12 b). are computed from the FSR values using Eqn. 9.

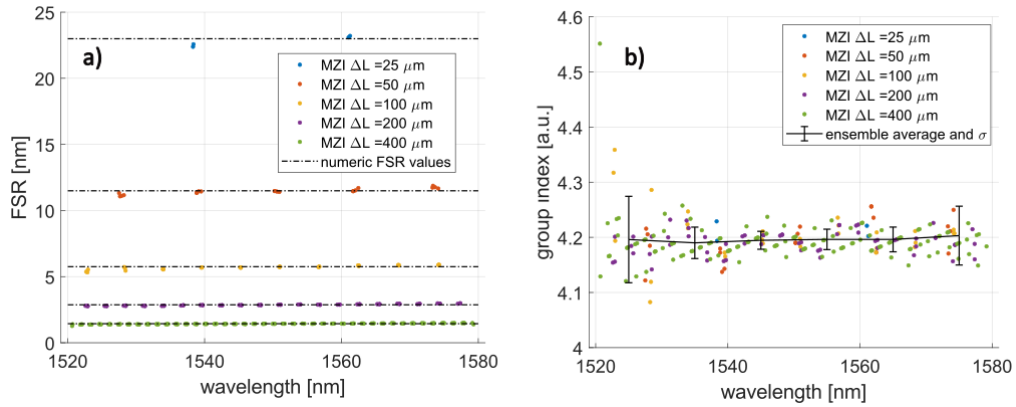


Fig. 12. a) FSR values extracted from measured MZI transmission curves and b)  $n_g$  values computed from extracted FSR values.

195 For an accurate estimation of  $n_{\text{eff}}$  and  $n_g$  at  $\lambda = 1550$  nm, the measured MZI transmission  
 196 spectra are fit with the following function:

$$F = 10 \log \left( \frac{1}{4} \left| 1 + \exp \left[ -i \frac{2\pi n_{\text{eff}} \Delta L}{\lambda} \right] \right|^2 \right) + b \quad (13)$$

where  $n_{\text{eff}}$  is expressed with Eqn. 10. The fit parameters are  $n_1$ ,  $n_2$  and  $n_3$  from Eqn. 10, and background scaling parameter  $b$  from 13. The parameter  $n_1$  is a non-dispersion corrected  $n_{\text{eff}}$  and is estimated by computing where Eqn. 6, the model for MZI transfer function, is zero, i.e. solving the following equation

$$-1 = \cos \left( \frac{2\pi n_{\text{eff}} \Delta L}{\lambda} \right) = \cos \left( \frac{2\pi n_1 \Delta L}{\lambda} \right) \quad (14)$$

where  $n_{\text{eff}} = n_1$  as the evaluation wavelength  $\lambda = \lambda_0$  and for an estimate of  $n_1$ ,  $\lambda_0$  is chosen as the wavelength closest to 1550 nm fullfilling Eqn. 14. An estimation of fit parameter  $n_2$  is obtained by inserting the  $n_g$  values obtained from the experimentally determined FSR values into Eqn. 11. An example of MZI transmission spectra fit is shown in Fig. 13.

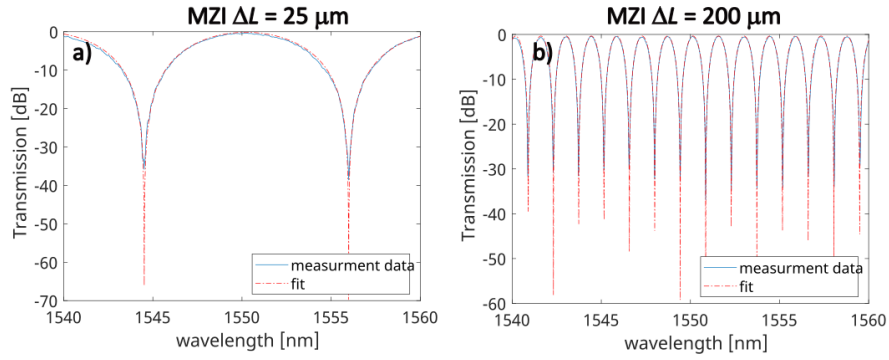


Fig. 13. Fitting to experimental transmission spectra from MZI with a)  $\Delta L = 25 \mu\text{m}$  and b)  $\Delta L = 200 \mu\text{m}$ .

The resulting values of  $n_{\text{eff}}$  and  $n_g$  from curve fitting are listed in table 2 with simulated values obtained from the corner analysis described in section 3.1.

Table 2. Comparison between  $n_{\text{eff}}$  and  $n_g$  as extracted from experiments and obtained from corner analysis simulation study.

|                  | experiment        | simulated min | simulated nom | simulated max |
|------------------|-------------------|---------------|---------------|---------------|
| $n_{\text{eff}}$ | $2.396 \pm 0.003$ | 2.370         | 2.440         | 2.466         |
| $n_g$            | $4.193 \pm 0.012$ | 4.162         | 4.180         | 4.239         |

The experimentally extracted values for  $n_{\text{eff}}$  and  $n_g$  fall well within the respective ranges as obtained from the corner analysis study shown good agreement between modeled MZI circuits and fabricated devices.

## 8. Conclusion

A full cycle of a typical design, modeling, layout and fabrication cycle of several MZI circuits has been reported. Waveguides were modeled within Lumerical MODE to simulate waveguide mode and model waveguide dispersion properties and yield  $n_{\text{eff}}$  and  $n_g$  values within an operational spectral bandwidth of 1500 nm - 1600 nm. This study was repeated for a corner analysis five waveguide geometries corresponding with manufacturing deviations from nominal waveguide width  $w = 500$  nm and thickness  $t = 220$  nm, thus incorporating fabrication tolerances. Compact models were made corresponding to each of the studied waveguide cross-sectional geometries for incorporation into circuit-level simulation studies of several proposed MZI circuits for manufacture. These MZI devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. and measured with a custom-built automated test setup yielding transmission spectra within the wavelength range of 1500 nm - 1600 nm. Analysis of the measured spectra yielded  $n_{\text{eff}}$  and  $n_g$  values of  $2.396 \pm 0.003$  and  $4.193 \pm 0.012$ , respectively. These values fall well within the expected ranges obtained from the corner analysis study, thus showing a good agreement between MZI circuit model and manufactured device.

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